

Comparing of feature selection algorithms

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Abstract—Feature selection plays a key role in data preprocessing in machine learning, and has an important impact on the accuracy and performance of algorithm prediction. This paper introduces the framework of feature selection and the generation of feature subsets. The feature selection methods are classified according to the correlation between feature selection and algorithm. Finally, the performance and accuracy of the feature generation algorithm are verified by experiments, and the corresponding conclusions are given.

Keywords—feature selection, Machine learning, evaluation criteria

I. INTRODUCTION

Machine learning has been widely used in all aspects. Feature selection, as a key link of data preprocessing in machine learning, becomes more and more important. In the aspects of image, gene and biological information, the sample data presents the characteristics of high-dimensional data. High-dimensional data brings great challenges to feature engineering. On the one hand, with the increase of data dimension, the data requirements for samples increase exponentially; On the other hand, the training cost is too high.

Dimension reduction methods can be divided into two categories. One is called feature extraction, which mainly converts sample data to low latitude features, that is, high latitude data into low latitude data through mathematical methods. The most typical method is principal component analysis [1]. Many studies have also proved that if there are not too many irrelevant dimensions in the features, The method of feature extraction can solve the problem of dimensional disaster to a certain extent [2]. However, because it is not the original data feature, the interpretability of the transformed new feature data meaning is relatively poor, and it is not received in many scenarios.

The other is feature selection, that is, selecting "effective" feature subsets from the original data features. The meaning of "effective" here includes several aspects: on the one hand, it has high correlation with the target results and can have a great influence on the prediction results; On the other hand, the feature correlation degree in the feature subset is low, which reduces the redundancy in the feature subset. The overall goal of feature selection is to reduce the number of features and dimensions without affecting the accuracy of prediction. In addition, feature selection is the processing in the features of the original data, which retains the semantics of the original data. In recent years, it has been widely concerned in many fields.

We compared the methods and ideas around the existing feature selection algorithms, compares the performance of the algorithms through experimental verification, and finally puts forward the difficulties and directions of the current research.

II. METHODS OF FEATURE SELECTION

The ultimate goal of feature selection is to obtain a feature subset. Ideally, the features in the feature subset should have high correlation with the prediction target and low correlation between features. Therefore, the task of feature selection is to select the features with high correlation on the one hand, and reduce the number of features without affecting the prediction on the other hand.

Feature selection methods can be divided into filter, wrapper and embedded.

A. Feature selection based on filtering

a) Evaluation of individual features

The overall idea of this method is to score each feature of candidate features under specific evaluation criteria, sort the features in descending order according to the score value, and directly select the first k features as the feature subset. Finally, the feature subset is used to train the learner to verify the advantages and disadvantages of the subset. For the evaluation of a single feature, the quality of evaluation criteria directly affects the quality of feature subsets. The existing research on evaluation criteria mainly focuses on distance, consistency, dependence, information measurement and so on.

Distance is mainly to evaluate the distance between features and categories [3]. The shorter the distance, the smaller the distance between features and categories, and the greater the correlation; It can also judge whether they are redundant according to the distance between features. The smaller the distance, the greater the redundancy. Distance is usually based on geometric distance, mainly including Euclidean distance, Ming distance, Mahalanobis distance, etc. The consistency standard judges the consistency of features and categories by means of proportion [4]. The feature subset obtained by this criterion is small, but it is only suitable for discrete features. It is generally only applied in classification problems. The dependency measurement uses statistical principles to evaluate the correlation between features and categories [5]; Unlike distance measurement and dependency measurement, which can only describe linear relationship, information measurement standard can measure the nonlinear relationship between features and between features and categories.

It can be seen that the evaluation of a single feature is mainly from spatial vector, statistics and information measurement. In terms of distance, the feature and analogy are represented by vectors, and then the distance between feature and vector is calculated; Using statistical theory to calculate the consistency of characteristics and classification; Information entropy, mutual information and information gain are used to measure information. For the evaluation of single feature, the relevant results of existing research are as follows:

Laplacian [6]

This score measures the discrimination of features, and the Laplacian score of feature f is defined as:

$$\text{Laplacian}(f) = \frac{\sum_{ij}(f_i - f_j)^2 s_{ij}}{\text{Var}(f)} \quad (1)$$

f_i represents the value of sample i on feature f ; s_{ij} represents the corresponding value in the weight matrix S . S is a simulated expression of the data spatial structure which describes the distance between two samples of the same kind (nearest neighbor). The bigger the distance between two samples of the same kind (nearest neighbor), the greater the corresponding weight, and vice versa; $\text{Var}(f)$ represents the variance of feature f . It is easy to know from the above formula that the lower the Laplacian score, the better the characteristic f .

In addition, constraint score and Fisher score also measure the importance of features from their ability to distinguish categories. Pearson correlation coefficient is a statistical method to calculate the linear correlation degree of two features. Pearson system can be used to judge the correlation degree between features and categories.

Pearson [7]

$$\mathbf{r}(f_i, f_j) = \frac{\text{Cov}(f_i, f_j)}{\sqrt{\text{Var}(f_i)\text{Var}(f_j)}} \quad (2)$$

$\text{Cov}(f_i, f_j)$ represents the covariance of feature f_i and f_j , and $\text{Var}(f_i)$ represents the variance of feature f_i . The features are scored by calculating the correlation coefficient between features and categories. The higher the score, the better the features. Pearson system can only calculate linear correlation, and more often there are complex nonlinear characteristics between features and categories.

MIC(Maximal Information Coefficient) [8]

MIC measures the degree of correlation between two variables x and y , the intensity of linearity or nonlinearity. According to the nature of MIC, MIC has universality and is not limited to specific function types when the sample size is large enough; This mic can be used not only for classification problems, but also for regression problems. Mic measures the correlation between features and categories (or interpreted features). The greater the value, the higher the correlation.

The calculation method of MIC is to discretize the relationship between two variables in two-dimensional space, and use the scatter diagram to represent it. Divide the current two-dimensional space into certain interval numbers in x and y directions, and then check the current scatter points falling into each grid. This is the calculation of joint probability. The formula is as follows:

$$\text{MIC}(X, Y) = \max_{|X||Y| < B} \frac{I[X; Y]}{\log_2(\min(|X|, |Y|))} \quad (3)$$

X indicates that the value field of x is divided into segment x , y indicates that the value field of y is divided into segment y , $|x| |y| < B$ indicates that the number of grids cannot be greater than B , and the numerator indicates different $x \times Y$ The maximum mutual information value under y grid division, and the denominator means normalizing the maximum mutual

information value under different divisions.

b) Evaluation for the entire feature subset

Another idea is not only to select subsets by sorting, but to evaluate the selected feature subsets, that is, to evaluate the feature subsets. Here, we need to pay attention to the generation process of feature subsets. Two aspects need to be considered in the process of generating feature subsets. One is the starting point, which refers to where to add feature subsets; There is also a search strategy, that is, how to traverse the original feature set to generate the best feature subset.

From the original search method based on feature subset, the search strategy can be divided into global optimal search, sequential search and random search. Sequential search and random search are collectively referred to as heuristic search strategies, including four search strategies corresponding to four search starting points: forward, backward, two-way and random search.

The method of enumeration comparison is the most intuitive way to obtain the optimal connection, but the efficiency is low. Heuristic search strategy is an important branch of optimization research. Its essence is to add some characteristics in the search process to make the search move towards the optimal direction effectively. Its efficiency is much better than the global optimal search, but generally only the local optimal solution can be obtained. Under the search strategy, each feature selected will evaluate the selected feature subset, The redundancy of feature subsets is reduced to a certain extent, and the efficiency is relatively high. The relevant scoring criteria mainly include correlation based feature selection (CFS) algorithm, which uses the merit score to evaluate the candidate feature subset. If the merit score is reduced by adding this feature, the feature can be eliminated.

CFS estimates the performance of feature subsets rather than single features. It introduces corresponding search strategies to select strongly related non redundant features. The idea is: good feature subsets contain features highly related to classes, but the features are not related to each other. The evaluation function introduced by CFS is:

$$\text{Merit}_s = \frac{k r_{fc}}{\sqrt{k + k(k-1) r_{ff}}} \quad (4)$$

Merit represents the category distinguishing ability of feature subset S , k represents the number of features in the selected feature subset S , r_{fc} represents the average correlation coefficient between feature f and category c and r_{ff} represents the average redundancy between feature f and other features in S . The higher the merit value, the better the performance of the feature subset S .

B. Feature selection based on Wrapper

The wrapper feature selection is different from the filter method. The filter method is independent of the learner, while the package method depends on the selected learning algorithm. The evaluation criterion of the feature subset adopts the learner to be used to judge the advantages and disadvantages of the feature subset. The selection of learning algorithms is not limited. Different algorithms can be selected in different scenarios. Support vector machine and k-nearest neighbor can be used for classification problems. Support vector machine is suitable for high-dimensional and small samples, and k-nearest neighbor is suitable for small sample

size and low dimension; Least squares regression, partial least squares regression (PLS), Lasso, etc. can be selected for the regression problem. If the data sample size is more than the feature number, least squares regression can be used to realize the regression modeling of multi independent variables on multi dependent variables and high-dimensional small sample data. Lasso can also be used for high-dimensional small sample data, and it has the function of feature selection.

After determining the learner, the selected features are directly used to train the learner in the feature selection process. The performance of the learner on the test set is used as the evaluation standard of the advantages and disadvantages of the features, and the encapsulated feature can select the features more suitable for the specific learning algorithm. For each candidate subset, the classifier needs to be retrained, so the speed of this method is slower than the filtering method. The advantage is that the scale of the selected optimized feature subset is relatively much smaller. In the specific selection process, the search method is also involved. The wrapper method based on heuristic search strategy is a more practical mainstream method of feature selection.

C. Feature selection based on Embedded

The so-called embedded feature selection is to fit the data through some special models, then take some evaluation attributes of the model itself as the evaluation index, and finally use the wrapped feature selection method to select. The embedded feature selection method can solve the problem of too high redundancy of filter algorithm based on feature sorting and too high time complexity of wrapper algorithm. It is a compromise between filter and wrapper.

The most commonly used models for embedded feature selection: decision tree model and model with regular items. There are least squares regression methods Lasso, mean fraction Lasso, k-split lasso based on L1 regular term. The nature of L1 regular term will shrink the regression coefficient towards 0, and the smaller coefficient may be compressed to 0. Therefore, the characteristics corresponding to those coefficients that are not 0 are the final feature subset. Classification decision tree is a classic embedded feature selection algorithm. The feature selection including ID3 and C4.5. Cart and other algorithms, the features used in training are the results of feature selection, which can be applied to classification and regression problems.

III. EXPERIMENTS

In order to compare the effects of different feature selection methods, we use three methods: filtering, wrapper and embedded.

We focus on the similarity of the features selected under different methods, the accuracy of the results under the features, and the running time of the selected features. For all data sets, we chose random forest, as shown in table I and figure 1.

TABLE I. TEST DATA SETS

Name	feature	specimen	validation set	test set
Divorce	54	170	68	102
Audit	18	777	310	467
Loan	26	10000	4000	6000

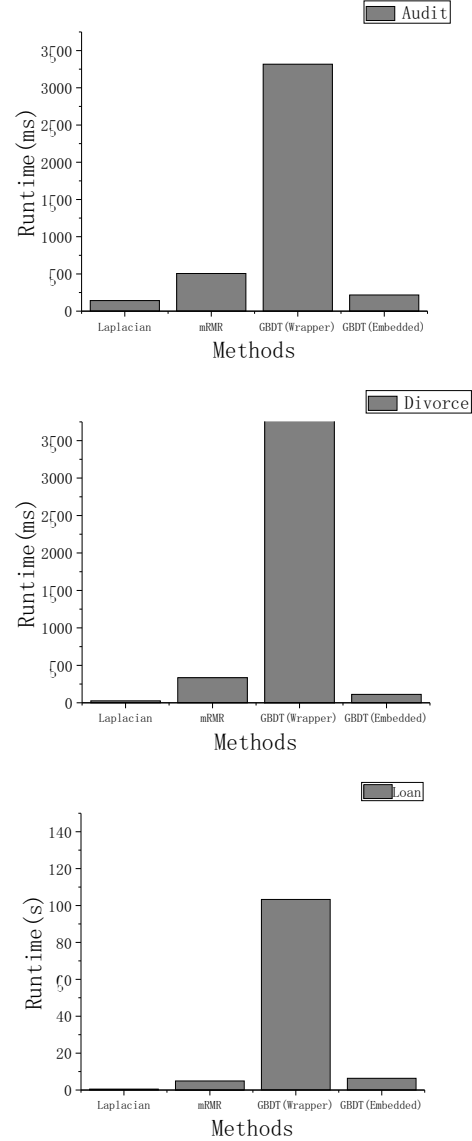


Fig.1 Runtime of datasets (Audit, Divorce, Loan)

From the experimental results, it can be seen that the time division of feature selection under wrapper is the highest. At the same time, with the increase of the number of samples, the time of wrapper feature selection will also increase significantly, and the increase of filter and embedded will be small.

From the results of similarity comparison, as shown in table II, it can be seen that the similarity of feature selection under encapsulation and embedded is high, and the similarity is low due to different sorting rules under filtering.

TABLE II. SIMILARITY COMPARISON

Dataset	Similarity(Filter)	Similarity(Wrapper&Embedded)
Loan	0.8	1
Divorce	0.7	0.8
Loan	0.5	0.9

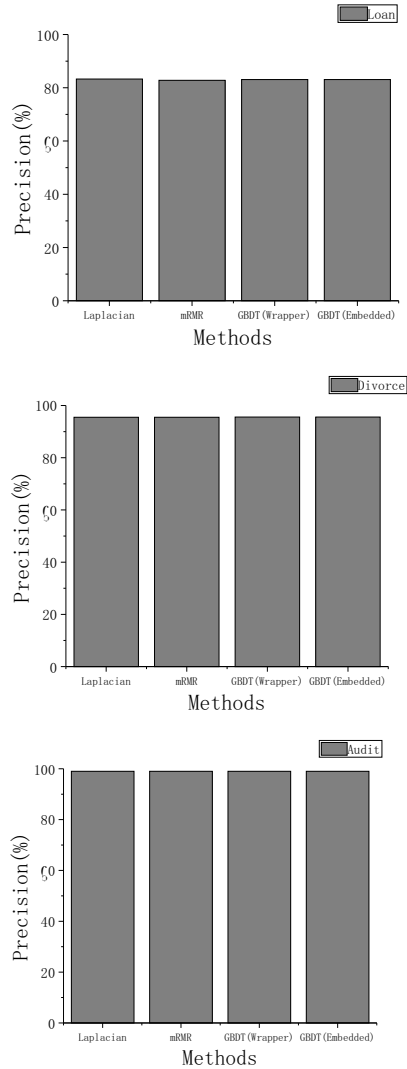


Fig.2 Precision of datasets (Audit, Divorce, Loan)

In terms of accuracy, as shown in figure 2, it can be seen that there is little difference in the final accuracy obtained by using the same learner after feature selection. In practice, this needs to further study the significance of feature selection.

IV. CONCLUSION

We directly compared the methods of feature selection. We can see from the text that feature selection has developed more mature, and there are relevant studies on different methods. No matter which method, its basic idea is to solve the correlation between features or feature subsets and results. Under the background of increasing computing power and less dimensional data, accuracy will become more important. The feature selection of high-dimensional data is more suitable to adopt the comprehensive method. For example, first use the filter to obtain a single feature with high correlation, and then determine the feature subset according to the correlation between the result and the feature. The research on the application of this kind of problem will be the trend in the future.

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REFERENCES

- [1] Liu C.Gabor- based kernel PCA with fractional power polynomial models for face recognition[J].IEEE Transactions on Pattern Analysis and Machine Intelligence, 2004, 26(5):572-581
- [2] Blum A L, Langley P.Selection of relevant features and examples in machine learning[J].Artificial Intelligence, 1997, 97(1/2):245-271.
- [3] Kira K, Rendell L A.A practical approach to feature selection[C]//Proceedings of the Ninth International Workshop on Machine Learning(ML 1992), Aberdeen, Scotland, UK, July 1-3, 1992.[S.l.]:Morgan Kaufmann Publishers Inc, 1992:249-256.
- [4] Lu, J., Ling, T.W., Chan, C.Y., Chen, T.: From region encoding to extended dewey: On efficient processing of xml twig pattern matching. In: VLDB. (2005) 193-204.
- [5] Arauzo-Azofra A, Benitez J M, Castro J L.Consistency measures for feature selection[J].Journal of Intelligent Information System, 2008, 30:273-292
- [6] Hall M A.Correlation- based feature subset selection for machine learning[D].Hamilton, New Zealand:University of Waikato, 1999.
- [7] He X, Cai D, Niyogi P.Laplacian score for feature selection[C]//Advances in Neural Information Processing Systems, 2006:507-513.
- [8] van't Veer L J, Dai H, van de Vijver M J, et al.Gene expression profiling predicts clinical outcome of breast cancer[J].Nature, 2002, 415(6871):530-536
- [9] Reshef D N, Reshef Y A, Finucane H K, et al.Detecting novel associations in large data sets[J].Science, 2011, 334(6062)